Batch: A2	Roll No.: 16010324044
Experiment / assignment / tutorial No. 8	
Grade: AA / AB / BB / BC / CC / CD /DD	
Signature of the Staff In-charge with date	

TITLE: Configure File Transfer Protocol using Cisco Packet Tracer.

AIM: To configure the FTP for given network topology using Cisco Packet Tracer.

OUTCOME: Student will be able to Understand operations of various application layer protocols

Theory:

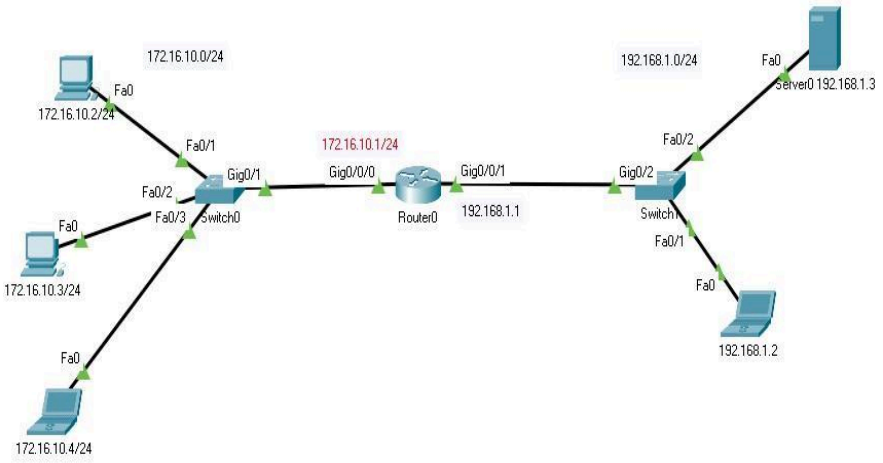
File Transfer Protocol (FTP) is an application layer protocol which moves files between local and remote file systems. It runs on the top of TCP, like HTTP. To transfer a file, 2 TCP connections are used by FTP in parallel: control connection and data connection.
FTP is a client/server protocol used for transferring files to or exchanging files with a host computer. It may be authenticated with user names and passwords. Anonymous FTP allows users to access files, programs and other data from the Internet without the need for a user ID or password. Web sites are sometimes designed to allow users to use 'anonymous' or 'guest' as a user ID and an email address for a password. Publicly available flies are often found in a directory called pub and can be easily FTPed to a user's computer. FTP is also the Internet standard for moving or transferring files from one computer to another using TCP or IP networks.

The File Transfer Protocol (FTP) is a standard network protocol used for the transfer of computer files from a server to a client on a computer network. FTP users may authenticate themselves with a clear-text sign-in protocol, normally in the form of a username and password, but can connect anonymously if the server is configured to allow it. For secure transmission that protects the username and password, and encrypts the content, FTP is often secured with SSL/TLS (FTPS) or replaced with SSH File Transfer Protocol (SFTP).

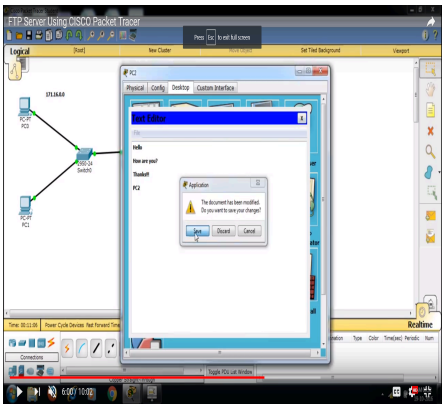
The first FTP client applications were command-line programs developed before operating systems had graphical user interfaces, and are still shipped with most Windows, Unix, and Linux operating systems. Many FTP clients and automation utilities have since been developed for desktops, servers, mobile devices, and hardware, and FTP has been incorporated into productivity applications, such as HTML editors.

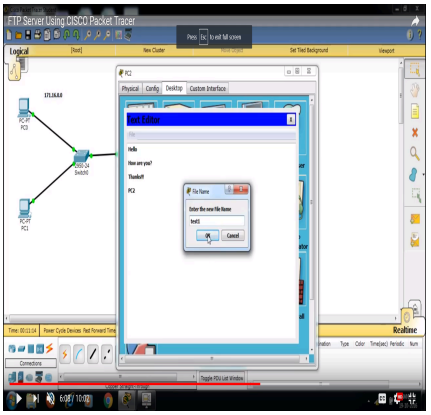
Procedure:

1. To begin with the experiment, open Cisco Packet Tracer
2. Create the lab and configure the basic IP addressing for all PCs and server for the network topology shown below:



3. Create a file in the laptop 192.168.1.2 as shown in the example below
For example:

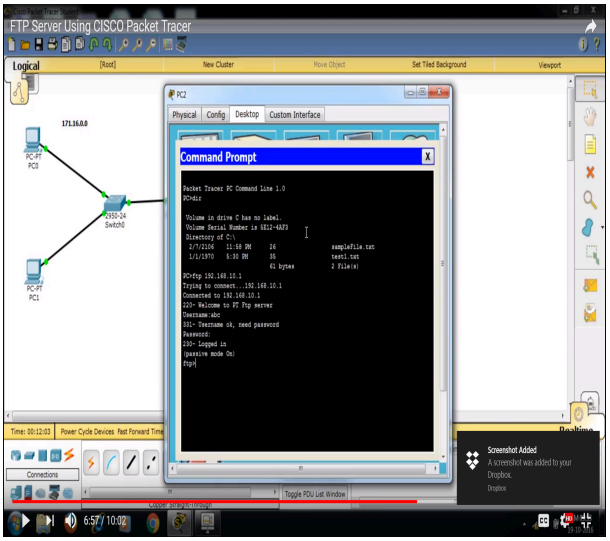




4. Save the file in the laptop directory as shown above

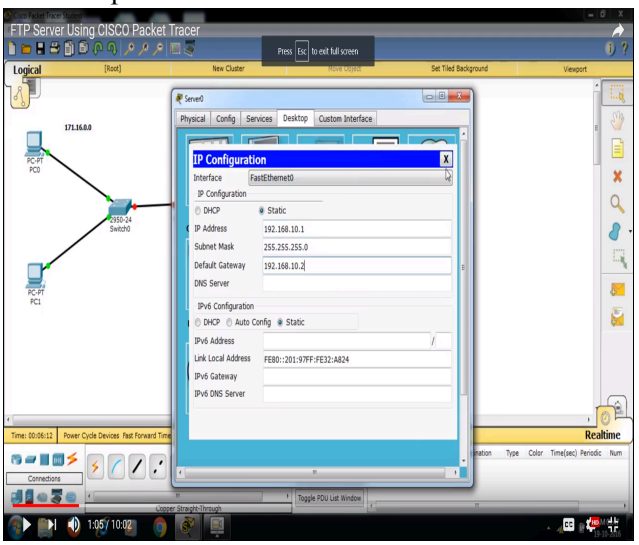
5.Check the directory of laptop

For example: go to command prompt and type dir and press enter



6. Go to server 0 and configure the IP address 192.168.1.3 and default gateway 192.168.1.1 in the server

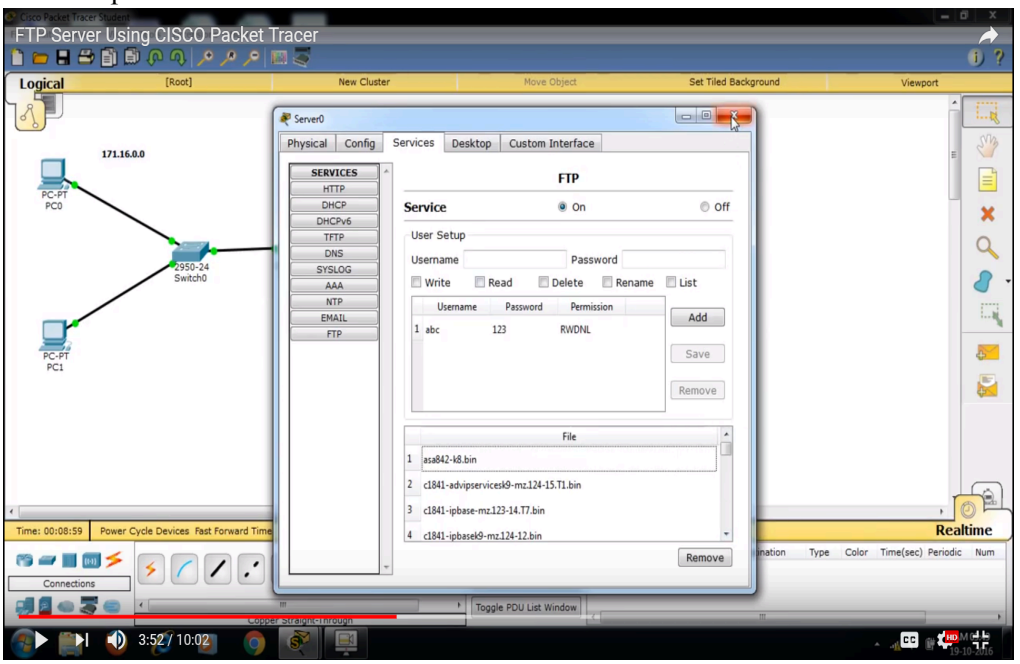
For example:



7. Go to the services in the server and make FTP services enabled from FTP server

8.Configure username ABC and password 123 and give all rights for read, write, delete, rename and list

For example:

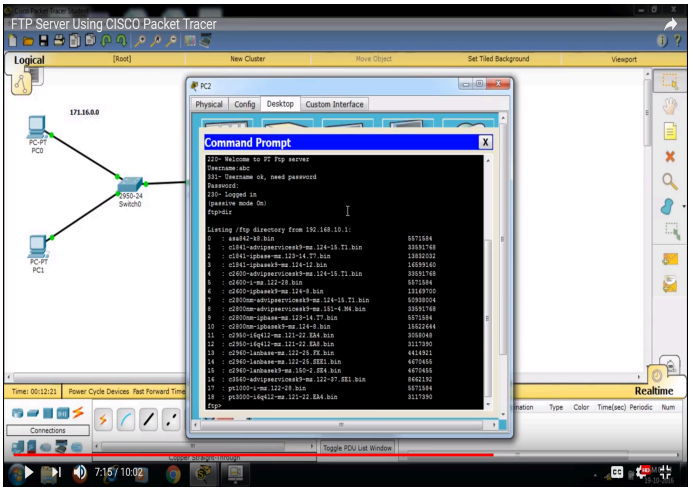


9. Click ADD to save the configuration on the server

10. Log on to ftp server using username and password from the laptop. Put the correct credentials to enter the configuration from the laptop.

Go to Laptop -> Command Prompt->ftp 192.168.1.3

Check the contents of the ftp server before saving the file to the server

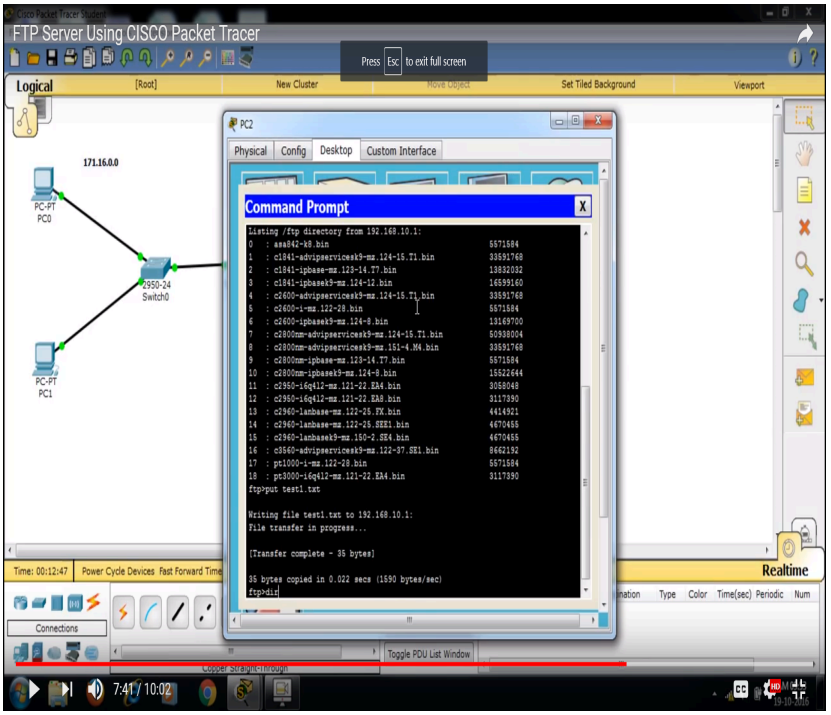


11. Now click on Laptop to save the file to the server

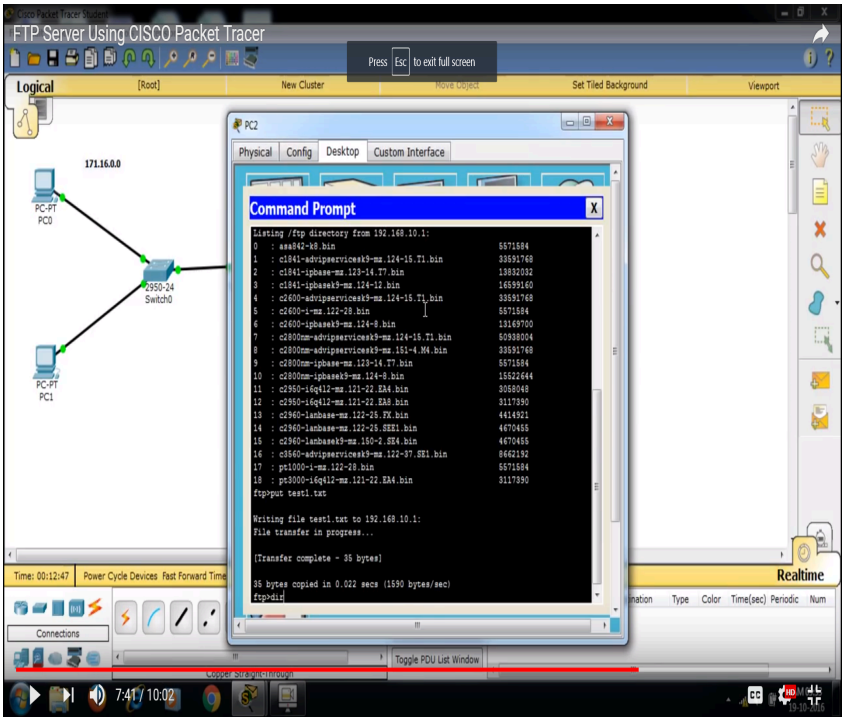
10. Go to Laptop -> Command Prompt->ftp 192.168.1.3

It will ask for username and password. Use put command to upload test1.txt

For example:



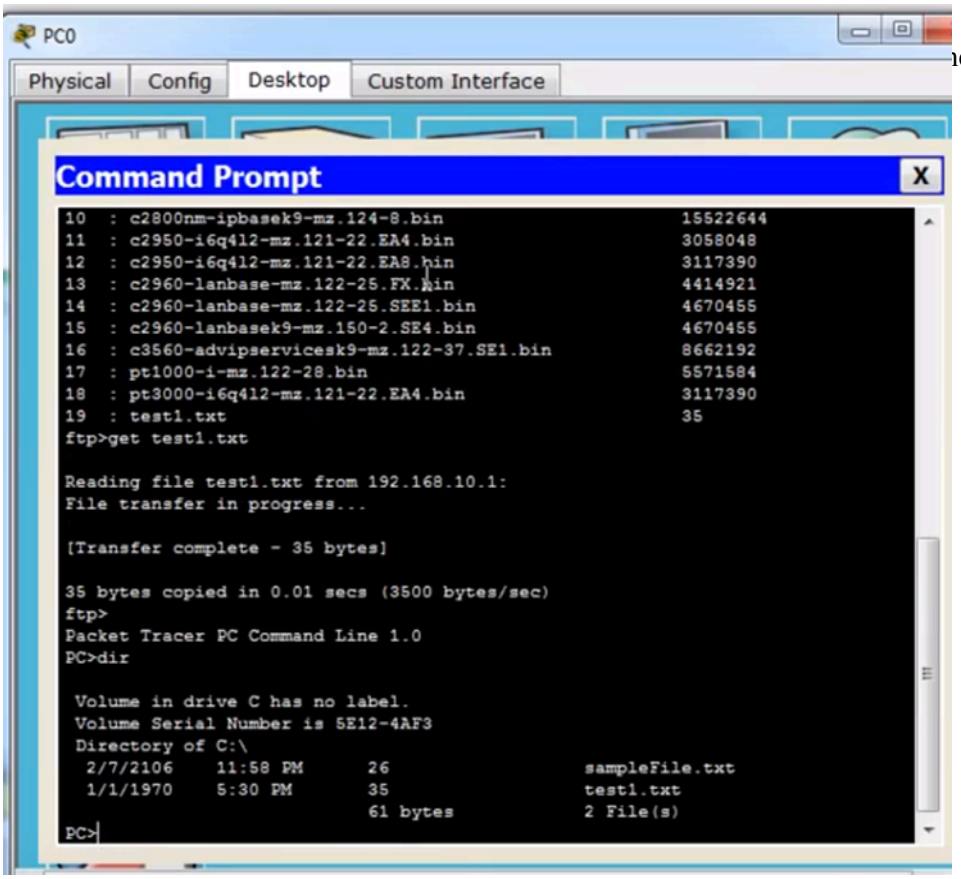
12. Check directory again to see test1 uploaded on server:



13. Now the file is on the server so we can access it from any other system as shown:

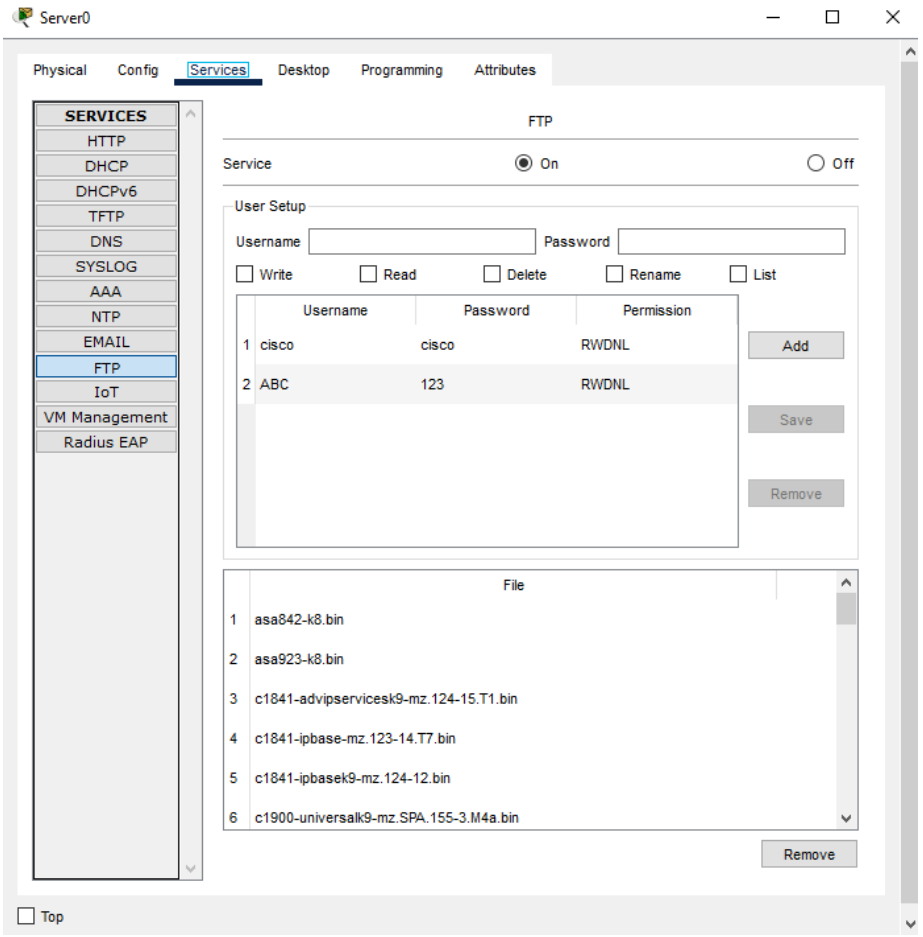
So first check the directory of PC0 where test1.txt is originally not present

Then we use the get command to get that file and again check the directory.



ıe file is copied

Output:


Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
FTP
Service
On
Off
User Setup
Username
Password
Write
Read
Delete
Rename
List
Username
Password
Permission
1 cisco cisco RWDNL
2 ABC 123 RWDNL
Add
Save
Remove
File
1 asa842-k8.bin
2 asa923-k8.bin
3 c1841-advipservicesk9-mz.124-15.T1.bin
4 c1841-ipbase-mz.123-14.T7.bin
5 c1841-ipbasek9-mz.124-12.bin
6 c1900-universalk9-mz.SPA.155-3.M4a.bin
Remove
Top

```
C:\>ftp 192.168.1.3
Trying to connect...192.168.1.3
Connected to 192.168.1.3
220- Welcome to PT Ftp server
Username:ABC
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.1.3:
0    : asa842-k8.bin                                5571584
1    : asa923-k8.bin                                30468096
2    : c1841-advipservicesk9-mz.124-15.T1.bin       33591768
3    : c1841-ipbase-mz.123-14.T7.bin                13832032
4    : c1841-ipbasek9-mz.124-12.bin                 16599160
5    : c1900-universalk9-mz.SPA.155-3.M4a.bin       33591768
6    : c2600-advipservicesk9-mz.124-15.T1.bin       33591768
7    : c2600-i-mz.122-28.bin                        5571584
8    : c2600-ipbasek9-mz.124-8.bin                  13169700
9    : c2800nm-advipservicesk9-mz.124-15.T1.bin     50938004
10   : c2800nm-advipservicesk9-mz.151-4.M4.bin      33591768
11   : c2800nm-ipbase-mz.123-14.T7.bin              5571584
12   : c2800nm-ipbasek9-mz.124-8.bin                15522644
13   : c2900-universalk9-mz.SPA.155-3.M4a.bin       33591768
14   : c2950-i6q412-mz.121-22.EA4.bin               3058048
15   : c2950-i6q412-mz.121-22.EA8.bin               3117390
16   : c2960-lanbase-mz.122-25.FX.bin               4414921
17   : c2960-lanbase-mz.122-25.SEE1.bin             4670455
18   : c2960-lanbasek9-mz.150-2.SE4.bin             4670455
19   : c3560-advipservicesk9-mz.122-37.SE1.bin      8662192
20   : c3560-advipservicesk9-mz.122-46.SE.bin       10713279
21   : c800-universalk9-mz.SPA.152-4.M4.bin         33591768
22   : c800-universalk9-mz.SPA.154-3.M6a.bin        83029236
23   : cat3k_caa-universalk9.16.03.02.SPA.bin       505532849
24   : cgr1000-universalk9-mz.SPA.154-2.CG          159487552
25   : cgr1000-universalk9-mz.SPA.156-3.CG          184530138
26   : ir800-universalk9-bundle.SPA.156-3.M.bin     160968869
27   : ir800-universalk9-mz.SPA.155-3.M             61750062
28   : ir800-universalk9-mz.SPA.156-3.M             63753767
29   : ir800_yocto-1.7.2.tar                        2877440
30   : ir800_yocto-1.7.2_python-2.7.3.tar           6912000
31   : pt1000-i-mz.122-28.bin                       5571584
32   : pt3000-i6q412-mz.121-22.EA4.bin              3117390
```

```
ftp>put hello.txt

Writing file hello.txt to 192.168.1.3:
File transfer in progress...

[Transfer complete - 17 bytes]

17 bytes copied in 0.091 secs (186 bytes/sec)
ftp>
```

```
C:\>dir

 Volume in drive C has no label.
 Volume Serial Number is 5E12-4AF3
 Directory of C:\

12/31/1969  16:0 PM            26        sampleFile.txt
               26 bytes              1 File(s)
```

```
C:\>ftp 192.168.1.3
Trying to connect...192.168.1.3
Connected to 192.168.1.3
220- Welcome to PT Ftp server
Username:ABC
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>get hello.txt

Reading file hello.txt from 192.168.1.3:
File transfer in progress...

[Transfer complete - 17 bytes]

17 bytes copied in 0.01 secs (1700 bytes/sec)
```

```
ftp>dir

Listing /ftp directory from 192.168.1.3:
0    : asa842-k8.bin                                5571584
1    : asa923-k8.bin                                30468096
2    : c1841-advipservicesk9-mz.124-15.T1.bin       33591768
3    : c1841-ipbase-mz.123-14.T7.bin                13832032
4    : c1841-ipbasek9-mz.124-12.bin                 16599160
5    : c1900-universalk9-mz.SPA.155-3.M4a.bin       33591768
6    : c2600-advipservicesk9-mz.124-15.T1.bin       33591768
7    : c2600-i-mz.122-28.bin                        5571584
8    : c2600-ipbasek9-mz.124-8.bin                  13169700
9    : c2800nm-advipservicesk9-mz.124-15.T1.bin     50938004
10   : c2800nm-advipservicesk9-mz.151-4.M4.bin      33591768
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14   : c2950-i6q412-mz.121-22.EA4.bin               3058048
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19   : c3560-advipservicesk9-mz.122-37.SE1.bin      8662192
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22   : c800-universalk9-mz.SPA.154-3.M6a.bin        83029236
23   : cat3k_caa-universalk9.16.03.02.SPA.bin       505532849
24   : cgr1000-universalk9-mz.SPA.154-2.CG          159487552
25   : cgr1000-universalk9-mz.SPA.156-3.CG          184530138
26   : hello.txt                                    17
27   : ir800-universalk9-bundle.SPA.156-3.M.bin     160968869
28   : ir800-universalk9-mz.SPA.155-3.M             61750062
29   : ir800-universalk9-mz.SPA.156-3.M             63753767
30   : ir800_yocto-1.7.2.tar                        2877440
31   : ir800_yocto-1.7.2_python-2.7.3.tar           6912000
32   : pt1000-i-mz.122-28.bin                       5571584
33   : pt3000-i6q412-mz.121-22.EA4.bin              3117390
ftp>
```

CONCLUSION

In this experiment, I successfully configured and implemented FTP using Cisco Packet Tracer. An FTP server with IP address 192.168.1.3 was configured and FTP services were enabled with appropriate user authentication and permissions. The file `test1.txt` was successfully uploaded from the laptop to the FTP server using the `put` command and subsequently downloaded to another system using the `get` command. Thus, the experiment successfully demonstrated file transfer between clients and an FTP server over a TCP/IP network and helped understand the working of FTP as an application-layer protocol.

Signature of faculty in-charge